

4 Models RFE Selected Feature on 7 Classification Models

RFE selects 'n' features separately using Logistic, SVM, Random Forest, and Decision Tree; then each selected-feature set is given to all seven Classifiers, and the resulting accuracies are arranged in the final DataFrame so that rows represent the RFE feature selector and columns represent the classification algorithm.

Step 1 - Original Dataset (indep_x, dep_y)

Step 2 – rfeFeature() → Select 'n' features using RFE – for all the four models → Logistic RFE features | SVM RFE features | Random RFE features | Decision Tree RFE features → rfelist stores the features

Step 3 – rfelist → for i in rfelist → splitScaler() → Train/Test split → StandardScaler → Logistic | SVML | SVMNL | KNN | Naïve | Decision Tree | Random Forest → Calculate Accuracy → Store Accuracy in lists

Step 4 – rfe_Classification() → Final DataFrame → showing Accuracy comparison of four models(Logistic, svc, Random and DecisionTree) vs 7 Classification models (Logistic, SVML, SVMNL, KNN, Naïve, Decision and Random)

Note: We apply Logistic RFE selected features on Logistic, SVM, SVML, SVMNL, KNN, Navie, Decision and Random in the first row; In the second row, we apply svc RFE selected features on Logistic, SVM, SVML, SVMNL, KNN, Navie, Decision and Random; similarly for the 3rd and 4th rows.

Outcome:

In this document, we will use the final dataframe to compare different 'k' values in four models RFE selected features vs 7 classification models with Accuracy value from confusion matrix.

1. K=3

Comparison of each Model's Accuracy values

```
[4]: result_rfe_df
```

[4]:	Logistic	SVML	SVMNL	KNN	Navie	Decision	Random
Logistic	0.94	0.94	0.94	0.94	0.94	0.94	0.94
svc	0.87	0.87	0.87	0.87	0.87	0.87	0.87
Random	0.94	0.94	0.94	0.94	0.9	0.91	0.92
DecisionTree	0.98	0.98	0.98	0.98	0.79	0.97	0.97

Best Model – Decision Tree for K=3

2. K=4

Comparison of each Model's Accuracy values

```
[11]: result_rfe_df
```

[11]:	Logistic	SVML	SVMNL	KNN	Navie	Decision	Random
Logistic	0.95	0.95	0.95	0.95	0.95	0.95	0.95
svc	0.96	0.96	0.96	0.96	0.96	0.96	0.96
Random	0.97	0.97	0.97	0.98	0.87	0.95	0.97
DecisionTree	0.98	0.98	0.92	0.98	0.81	0.98	0.98

Best Model – Decision Tree for K=4

3. K=5

Comparison of each Model's Accuracy values

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[21]: result_rfe_df
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	Logistic	SVML	SVMNL	KNN	Navie	Decision	Random
Logistic	0.98	0.98	0.98	0.98	0.98	0.98	0.98
svc	0.99	0.99	0.99	0.99	0.99	0.99	0.99
Random	0.97	0.97	0.98	0.97	0.91	0.96	0.98
DecisionTree	0.95	0.98	0.93	0.94	0.85	0.97	0.98

▼ Columns Selected for k=5

Best Model – svc, k=5

4. K=6

Comparison of each Model's Accuracy values

```
[11]: result_rfe_df
```

	Logistic	SVML	SVMNL	KNN	Navie	Decision	Random
Logistic	0.98	0.98	0.98	0.96	0.98	0.98	0.98
svc	0.99	0.99	0.99	0.99	0.99	0.99	0.99
Random	0.98	0.98	0.99	0.96	0.92	0.95	0.98
DecisionTree	0.96	0.96	0.97	0.95	0.85	0.97	0.96

▼ Columns Selected for k=6

Best Model – svc, k=6

5. K=7

Comparison of each Model's Accuracy values

```
[11]: result_rfe_df
```

```
[11]:
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	Logistic	SVML	SVMNL	KNN	Navie	Decision	Random
Logistic	0.98	0.98	0.98	0.97	0.98	0.98	0.98
svc	0.99	0.99	0.99	0.99	0.99	1.0	0.99
Random	0.98	0.98	0.98	0.98	0.91	0.96	0.96
DecisionTree	0.95	0.96	0.96	0.94	0.88	0.97	0.95

Columns Selected for k=7

Best Model – svc, k=7

6. K=8

Comparison of each Model's Accuracy values

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[11]: result_rfe_df
```

```
[11]:
```

	Logistic	SVML	SVMNL	KNN	Navie	Decision	Random
Logistic	0.98	0.98	0.98	0.98	0.98	0.99	0.97
svc	0.99	0.99	0.99	0.98	0.99	1.0	0.99
Random	0.98	0.98	0.99	0.98	0.92	0.96	0.97
DecisionTree	0.96	0.97	0.97	0.98	0.88	0.95	0.97

Columns Selected for k=8

Best Model – svc, k=8

k value	Model Name of RFE Selected Features	Most Frequent Accuracy in 7 Classification Models (Highest in the Dataframe)	Frequency Of Occurrence	Comments
3	Decision Tree RFE Selected Features	0.98	4	
4	Decision Tree RFE Selected Features	0.98	5	
5	svc RFE Selected Features	0.99	7	All Models
6	svc RFE Selected Features	0.99	7	All Models
7	svc RFE Selected Features	0.99	6	Accuracy = 1 for DT model
8	svc RFE Selected Features	0.99	6	Accuracy = 1 for DT model

Interpretation:

From the above table, we can understand that 'svc RFE Selected Features' perform well in all the 7 Classification models (Logistic, SVML, SVMNL, KNN, Navie, DecisionTree and RandomForest) when k = 5, 6, 7 and 8.